



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A' grade (3.58/4) | Awarded Category – I by UGC

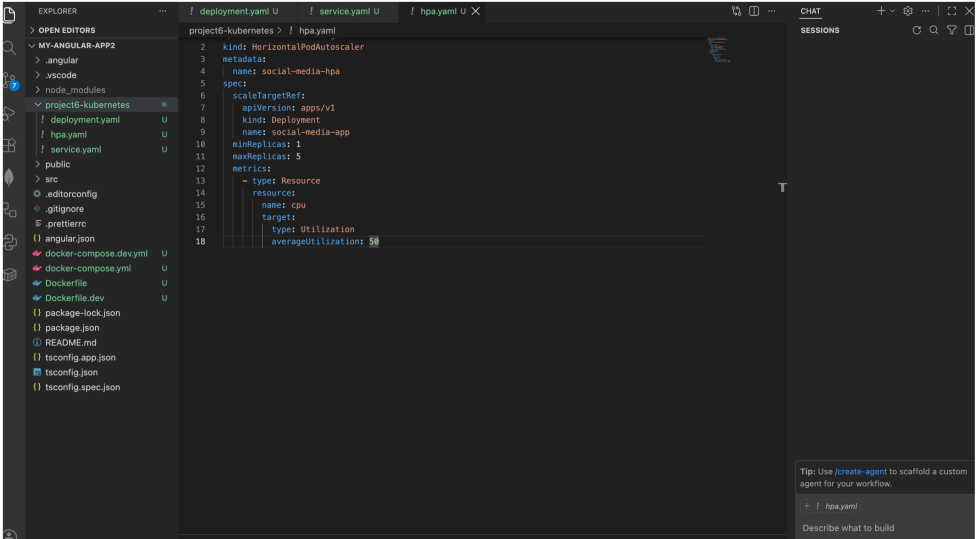
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 6

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	BOKHIT MAHAMAT TOM
PRN No.	23070122249
Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 Sept, 2026
Title of Assignment:	Social Media Underlying Infrastructure Challenges – Using Kubernetes
Theory:	Kubernetes is a container orchestration platform used to deploy, manage, and scale containerized applications. It helps applications remain available when the number of users or workload increases. In a social media application, the number of users and requests can change continuously. Running the application with only one Pod may not be sufficient during periods of high traffic. Kubernetes Horizontal Pod Autoscaler (HPA) provides automatic scalability by increasing or decreasing the number of Pods based on resource utilization such as CPU. In this project, a Kubernetes Deployment is used to manage the Social Media application, while a Service provides network access to the application. Metrics Server collects CPU and memory usage from the running Pods. The HPA monitors the CPU utilization and automatically adjusts the number of replicas according to the configured limits. The HPA is configured with a minimum of 1 replica and a maximum of 5 replicas, with a target CPU utilization of 50%. When the workload increases and CPU utilization exceeds the target, Kubernetes can create additional Pods. When the workload decreases, unnecessary Pods can be

removed.

Thus, Kubernetes autoscaling helps the Social Media application handle changing workloads while maintaining scalability and efficient resource utilization



```
2 kind: HorizontalPodAutoscaler
3 metadata:
4   name: social-media-hpa
5 spec:
6   scaleTargetRef:
7     apiVersion: apps/v1
8     kind: Deployment
9     name: social-media-app
10  minReplicas: 1
11  maxReplicas: 5
12  metrics:
13    - type: Resource
14      resource:
15        name: cpu
16        target:
17          type: Utilization
18          averageUtilization: 50
```

```
minikube Ready control-plane 25m v1.35.1
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 24m v1.35.1
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get pods
NAME READY STATUS RESTARTS AGE
load-generator 1/1 Running 0 17m
social-media-app-c4ff55b6c-mjzsp 1/1 Running 0 24m
social-media-app-c4ff55b6c-wmsnp 1/1 Running 0 15m
○ bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes %
```

```
social-media-app-c4ff55b6c-wmsnp 1/1 Running 0 15m
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get services
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S)
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP
social-media-service NodePort 10.105.27.221 <none> 80:30999/TCP
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get pods
NAME READY STATUS RESTARTS AGE
load-generator 1/1 Running 0 18m
social-media-app-c4ff55b6c-mjzsp 1/1 Running 0 25m
social-media-app-c4ff55b6c-wmsnp 1/1 Running 0 16m
○ bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes %
```

```

bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get nodes
minikube   Ready    control-plane   24m   v1.35.1
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
load-generator                      1/1     Running   0           17m
social-media-app-c4ff55b6c-mjzsp    1/1     Running   0           24m
social-media-app-c4ff55b6c-wmsnp     1/1     Running   0           15m
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP          26m
social-media-service NodePort    10.105.27.221 <none>         80:30999/TCP     24m
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
load-generator                      1/1     Running   0           18m
social-media-app-c4ff55b6c-mjzsp    1/1     Running   0           25m
social-media-app-c4ff55b6c-wmsnp     1/1     Running   0           16m
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get hpa
NAME                REFERENCE                               TARGETS   MINPODS   MAXPODS   REPLICAS   AGE
social-media-hpa    Deployment/social-media-app              cpu: 42%/50%  1         5         2          20m
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
load-generator                      1/1     Running   0           20m
social-media-app-c4ff55b6c-mjzsp    1/1     Running   0           26m
social-media-app-c4ff55b6c-wmsnp     1/1     Running   0           18m
• bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % kubectl get hpa
NAME                REFERENCE                               TARGETS   MINPODS   MAXPODS   REPLICAS   AGE
social-media-hpa    Deployment/social-media-app              cpu: 41%/50%  1         5         2          21m
○ bokhitmahamattom@Bokhits-MacBook-Air project6-kubernetes % 

```

Subject Faculty Name: Prof. Arijit Dutta